AIRBUS PROBLEM STATEMENETS

AUTONOMOUS SG 2026 | ACACIA HACKATHON SOCIETY X NUS PRODUCT CLUB

DATASETS: https://drive.google.com/drive/u/1/folders/13h0JZnSqhYTX3rV8dGDGy5AiwaADQvJR

Challenge #1

" How could we leverage AI-driven technologies for satellite collision avoidance "

Description

With low earth orbit (LEO) being crowded with satellites and space debris, satellite operators receive thousands of data messages daily warning of potential close encounters. Evaluating every encounter and determining if corrective action is required necessitates too much manual labor and is vulnerable to human error.

Want to see

A **collision risk evaluator** that reads orbital conjunction telemetry and subsequently calculates the collision probability (Pc) for satellite-debris pairs, automatically flagging high-risk events that require maneuver planning.

Risk classification: Algorithm predicting whether an event is high risk, requires maneuvering, or safe (backed by probability)

In accordance with *NASA Spacecraft Conjunction Assessment and Collision Avoidance Best Practices Handbook (NASA/SP-20205011318)* and *ESA Space Debris Office Standards*

Safe: $P_c < 10^{-6}$ and miss distance >5.0km

Elevated risk: $10^{-6} \leq P_c < 10^{-4}$ or miss distance between 1.0km and 5.0km

Critical (Maneuver Required): $P_c \geq 10^{-4}$ (i.e. risk >= -4.0 in the dataset) or miss distance <1.0km

Dashboard UI: Visual platform showing trajectories, collision risk scores and miss distances

Warnings, in 24 hour frame how many come within the defined dangerous distance, generate insights

Provided data

Collision Avoidance Challenge dataset (file in *Challenge #1\archive(1).zip*)

- Can use only columns **event_id, time_to_tca, miss_distance, relative_speed, relative_position_r, relative_position_t, relative_position_n , c_object_type, c_sigma_r, t_sigma_r, max_risk_estimate, F10, AP** for simplicity purposes
- In **archive(1).zip:**
- Use **train_data** to train models, fit parameters, relationships between telemetry features etc.
- Has column called *risk* -> represents log10(pc); map this to operational risk tiers to train and validate models

- Use **test_data** to generate predictions on rows, for outputs to be scored to see how well the model generalises unseen encounters

Challenge #2

" How might AI help air traffic controllers, pilots and ground operations staff make faster and better decisions? "

Description

Today, when a runway closes, weather delays occur or ground equipment fail unexpectedly, communications are reliant on human phone calls, manual software updates etc., leading to cascading delays, excessive holding patterns, and excessive fuel consumption. The future of Air Traffic Management relies on AI assistants that can autonomously communicate and negotiate trade-offs across organisational boundaries.

Want to see

A **Multi-Agent Communication Framework** where ATC AI agents, Pilot AI agents, and ground ops agents are able to negotiate resolutions to real-time disruptions without human intervention until a consensus is reached.

ATC agent → must take care of runway wake separation, taxiway blockage, maximise sector throughput

Pilot Agent → Preserve fuel reserves, ensure flight safety, minimise turbulence and protect passengers' connections

Ground Agent → Optimise gate utilisation, avoid ground-crew overtime, minimise turnaround delays

These 3 agents must exchange structured data, evaluate cross-domain constraints, and converge on an agreed solution. This includes sending **structured messages**, **explicit constraint checking, traceable consensus log**, **deterministic non-AI fallback**

What to avoid

- Combining 3 roles into a single prompt call; The point is to test multi-agent coordination across independent domains
- LLM agents talking to each other in endless, unconstrained loops with no convergence logic / turn limits
- Disregard of safety margins; AI agents should not be allowed to accept solutions that ignore operational constraints

Provided data

US Department of Transportation data for 2015 flight delays and cancellations

- File in *Challenge #2\DATA*
- Download this, pick any row with WEATHER_DELAY value of > 15, or LATE_AIRCRAFT_DELAY > 20
- Pass the parameters in that row into the 3 AI Agents

- Demonstrate how agents negotiate **revised arrival time, gate assignment, or flight plan adjustment** in under 3 turns

Challenge #3

" How might we optimise the SAF supply chain in Kalimantan? "

Description

There is a need to aggressively scale Sustainable Aviation Fuel (SAF) adoption to meet decarbonisation goals. Kalimantan (Indonesian Borneo) possesses abundant agricultural and forestry residues, making it a potential hub for regional SAF production. However, transporting raw biomass, establishing infrastructure, and delivering finished SAF to international airports requires strategic decision-making, and often comes with complex logistical and capital trade-offs.

Want to see

- A **SAF supply chain optimisation model** across Kalimantan that minimises the total cost (OpEx + CapEx + Logistics), while meeting SAF demand at Kalimantan's 4 international airports (Supadio Intl, Iskandar Intl, Sultan Aji Muhammad Sulaiman Sepinggan Intl, Juwata Intl)
- Should include optimal coordinates of depots & refineries
- Facility sizing decisions
- Material flow network
- Infrastructure placement: Deciding the spatial locations and sizes (Small vs Large) for pre-processing depots and SAF refineries
- Optimal routing of raw feedstock to depots, depots to refineries, finally refineries to airports

Feedstock quantity - **6800kt**

Facility Type	Size	Cost (USD)	Production Capacitykilotonne
Depot	Small	2.25M	150 kt
Depot	Large	4.5M	300 kt
Refinery	Small	480M	500 kt
Refinery	Large	765M	1087 kt

Truck Size	Farm to Depot	Cost (USD)tonne/km	Depot to Refinery	Cost (USD)tonne/km	Refinery to Airport	Cost(USD)tonne/km
Small	3 t	0.10/t/km	18 t	0.12/t/km	16 t	0.08/t/km
Large	9 t	0.10/t/km	30 t	0.12/t/km	32 t	0.08/t/km

Data provided

Airport locations

- Download data/airports.dat for precise airport locations (including Name, City, Country, Latitude, Longitude)
- **NOTE: ONLY FOUR AIRPORTS TO BE CONSIDERED: (Supadio Intl, Iskandar Intl, Sultan Aji Muhammad Sulaiman Sepinggan Intl, Juwata Intl)**
- File location: *Challenge #3 - SAF\Airport location\airport.dat*

Biomass locations

- 80 locations, with a range of **20 - 100** kilotons of biomass per location; total biomass is 6,800 kilotons
- File location: *Challenge #3 - SAF\Biomass locations\kalimantan_80_locations_6800kt_full_range.csv*

Challenge #4

" How might AI help people find, understand, and verify regulated technical information? "

Description

Sensitive regulatory documents include airworthiness directives, maintenance manuals and accident reports. Often, mechanics and engineers need to find these precise instructions quickly with full trust in the output. GenAI models often hallucinate or blend instructions across different types of equipment. This can lead to dangerous mishaps in operational situations.

Want to see

A **traceable & context-aware search engine** that ingests multi-page technical documents, parses exact applicability rules, answers natural language queries with citations. If the information is ambiguous, it should confidently refuse to answer instead of giving an unreliable answer.

Specific references → Every claim or sentence must link to its exact source metadata

Zero hallucination → System must not output hallucinated responses

Configuration filtering → System must methodically evaluate conditional applicability rules (e.g. *Directive A only applies to serial numbers S-1000 to S-1500*)

Revision Logic → User must be aware if the document has multiple versions, and what the latest version is

Clear communication of uncertainty → If prompt falls outside the scope of the data or contradicts existing constraints, or involves ambiguous terms, the system must clearly output a message that expresses its inability to complete the task

Sub 3-second retrieval latency → Satisfactory search performance across 1000+ pages

Want to avoid

LLM guessing → LLM attempting to fill knowledge gaps using its pre-trained weight memory

Unformatted data → Returned data must follow appropriate structure, cannot just be text randomly written

Provided data

Airworthiness Directives (ADs)

- File is in ***Challenge #4\432 entries of FAA Airworthiness Directives***

Challenge #5

" How might we leverage historical weather observation logs and flight performance datasets to predict airport runway capacity reductions and flight delay risks 3 to 12 hours before severe weather events occur? "

Description

Severe weather accounts for over 70% of all air traffic delays globally. When severe weather hits major airport hubs, air traffic controllers and airport operators must dynamically reduce hourly arrival and departure rates. Because capacity reductions are conducted reactively when weather actually arrives, aircraft end up stuck in costly airborne holding patterns or stranded on taxiways with engines running, wasting fuel and causing severe downstream flight cancellations.

In this challenge, participants must build a **Predictive Airport Capacity & Risk Model** that ingests publicly available weather observations (METARs/TAFs and NOAA storm logs) and historical flight performance data to forecast hourly runway arrival acceptance rates (AAR) and departure delay risks 3 to 12 hours in advance.

Want to see

Multi-source data integration → Merging airport weather observation logs with flight delay records using shared timestamps and airport station identifiers

Predictive Model Performance & Accuracy → Accurate prediction of Arrival Acceptance Rates, measured by Root Mean Squared Error

Delay Risk Classification → Specific classification of flights that face a severe delay risk (> 45 minutes)

Interactive dashboard & decision support for ATC → Dashboard that displays risk scores alongside historical trends, and recommendations for air traffic controllers

Automatic processing of public datasets → Scripts that download and process public datasets automatically without manual human intervention

Want to avoid

Data Leakage Across Time Steps → Using weather measurements recorded when flight arrives to predict delays; Instead use historical forecast/observation logs available ***prior*** to the event

Ignoring Runway Directionality → Evaluating raw wind speed without calculating the actual crosswind and tailwind components relative to specific airport runway orientations

Over-Fitting to a Single Airport → Building a model that only works on one specific airport layout rather than designing a generalized pipeline that accepts any airport ICAO code in the dataset

Evaluating Un-Normalized Delays → Comparing raw delay minutes between massive hub airports (e.g., Chicago O'Hare) and smaller regional airports without normalizing by overall flight volume

Provided data

Airport weather data from 2023 to 2025

- **Select only two airports: Atlanta Hartsfield-Jackson (ATL) & Dallas Fort-Worth (DFW)** ((Departing from, and arriving to); Weather telemetry included here, such as air temperature, precipitation, wind speed, wind gust peak, air pressure, snow depth, and weather condition nodes
- Assume the following:
- **Hartsfield-Jackson Atlanta International Airport**
- **All flights take-off and land at heading degree 80**
- **Dallas Fort Worth International Airport**
- **All flights take-off and land at heading degree 180**
- **File in** *Challenge #5\Flight_Data_With_Weather_Final.csv*

END OF DOCUMENT

Judging Rubric

AI & Aerospace Technology Track

Six criteria. Score each from 1 to 5; apply the weights below. Total weight: 100%.

Problem realism 20%	Technical soundness 25%	Honest evaluation 20%
Trust and traceability 20%	Impact 10%	Time discipline 5%

Problem realism

20% | 1-5

Does this solve a specific aerospace workflow problem?

Name the user, current process and cost in time, money or risk. Explain the constraints of the chosen challenge, rather than describing aviation in general.

STRONG
A credible user, narrow workflow and clear status quo.

WEAK
A generic aviation chatbot or a solution looking for a problem.

Technical soundness

25% | 1-5

Does the prototype work, with the right method for each task?

Show an end-to-end build. Use LLMs for reading and explanation; use solvers, rules or statistics for scheduling, constraints and detection. Test beyond a scripted demo.

STRONG
Working inputs and outputs; explicit constraints and a sound LLM / non-LLM split.

WEAK
LLM-only legality checks, opaque risk scores or a hard-coded demo.

Honest evaluation

20% | 1-5

What was tested, against what baseline, and where does it fail?

State metrics, test data and results. Label synthetic data and its limits. Show one case where the system is wrong, uncertain or declines, and explain how the user knows.

STRONG
A baseline, measured results and a visible failure with a reason.

WEAK
Only success cases, unsupported accuracy claims or hidden data limits.

Judging Rubric

AI & Aerospace Technology Track | Criteria and scoring guidance

Trust and traceability

20% | 1-5

Can a qualified person verify the output before acting?

Trace claims to a document, revision and section, a timestamped signal or an applied rule. Handle applicability and missing evidence. Keep the qualified person in control.

STRONG
Checkable sources, visible uncertainty and a clear refusal when needed.

WEAK
Confident unsourced answers or a claim to replace expert judgement.

Impact

10% | 1-5

What improves over the current process, and by how much?

Compare against the stated baseline: delay minutes, supplier impact, retrieval quality or false alarms, as relevant. Distinguish measured improvement from estimates.

STRONG
A credible before/after measure and realistic adoption constraints.

WEAK
Large benefit claims with no baseline or unstated assumptions.

Time discipline

5% | 1-5

Was the project submitted early and presented efficiently?

Earlier valid submission is better, using the recorded submission time. Also assess a clear, focused presentation completed within 3 minutes. Give one combined 1-5 score.

STRONG
Earlier submission and a focused, complete presentation within 3 minutes.

WEAK
Later submission and a rushed, unfocused or over-time presentation.

SCORING AND JUDGES' NOTES

Scale: 1 = weak; 2 = limited; 3 = credible; 4 = strong; 5 = exceptional evidence.

Weighted total / 100 = sum of (criterion score / 5 x criterion weight).
Example: technical soundness scored 4/5 earns 20 of its 25 points.

Score criteria separately. Ask: "Where does this fail, and how would your user know?"
Reward sound restraint, not presentation polish. Each project is evaluated by 3 judges.
Time is only 5%: earlier submission does not substitute for quality or evidence.